3D Seismic Tomography with Simultaneous Hypocenter Estimation via Expectation–Maximization: Method and Application

Egor Podoliako[1] Alexander Derendyaev[1]

[1]Institute for Information Transmission Problems, Moscow, Russia

Corresponding author: Egor Podoliako (podoliako.e@gmail.com)

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Abstract

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1 Introduction

Local-earthquake travel-time tomography uses picked seismic arrival times to estimate three-dimensional velocity structure and earthquake locations. Early work already formulated the inversion in terms of both source and medium parameters (Aki & Lee, 1976). An earthquake hypocenter is not an independently measured datum: its location is inferred from arrival times using an assumed velocity model, while velocity reconstruction in turn depends on the event locations (Kissling, 1988; Thurber, 1992). Treating catalog hypocenters as fixed inputs can therefore propagate location errors and dependencies on the models and procedures used to construct the catalog, particularly when those assumptions or location uncertainties are not documented. This coupling motivates estimating earthquake locations and velocity structure together.

Travel-time tomography remains useful for constructing and refining local velocity models (Fichtner et al., 2024). Such models can also provide starting points for waveform-based inversion, whose results may depend on the initial model (Virieux & Operto, 2009). There is therefore continued demand for practical methods that use arrival-time data to estimate three-dimensional structure without assuming accurately known earthquake locations.

Established joint-inversion methods iteratively update the velocity structure and an individual location estimate for each earthquake (Thurber, 1983). They differ in how those estimates are initialized: tomoDD uses initial event locations (Zhang & Thurber, 2003), whereas LOTOS determines them from arrival times and station coordinates in a one-dimensional reference model before three-dimensional inversion (Koulakov, 2009). In these formulations, locations enter velocity updates as individual estimates rather than as weighted alternatives. Probabilistic methods address different parts of this problem. Event-location methods can characterize alternative hypocenters given a prescribed velocity model (Lomax et al., 2000; Smith et al., 2022). Bayesian joint inversion has been demonstrated for both one-dimensional velocity structure (Ryberg & Haberland, 2019) and three-dimensional local-earthquake tomography using trans-dimensional Monte Carlo sampling (Piana Agostinetti et al., 2015); the latter jointly samples hypocenters and velocity structure without requiring catalog locations as starting coordinates. DeepGEM proposes generalized expectation–maximization for blind inversion (Gao et al., 2021),

while a recent three-dimensional variational double-difference method quantifies uncertainty in the velocity structure (Yang et al., 2025). These approaches differ in their treatment of location uncertainty and computational strategy; joint probabilistic 3D inversion without catalog starting locations is therefore not unique to EMTomo.

A practical motivation is to reconstruct three-dimensional velocity structure when reliable catalog hypocenters are unavailable, without committing the velocity update to a single estimated location for each event. We present EMTomo, which starts from arrival-time picks, station coordinates, and an initial velocity model, but does not require initial hypocenter coordinates. It alternates between evaluating possible event locations in the current model and updating the velocity model using either one selected location (hard assignment) or several weighted location hypotheses (soft assignment). This is a computationally different choice from the joint sampling of hypocenters and velocity structure by Piana Agostinetti et al. (2015); we do not claim that joint probabilistic 3D inversion without initial catalog locations is new. Our question is whether retaining several plausible locations is useful in this iterative travel-time inversion setting, and how the resulting reconstructions compare with an established joint-inversion workflow.

We investigate this question in three synthetic experiment series. First, checkerboard tests keep the inversion-cell size fixed while varying the size of the cubic velocity anomalies; they compare hard-assignment EMTomo with tomoDD initialized from hypocenters perturbed by location noise. Second, a layered velocity model with spherical anomalies, inverted from a layered starting model, is tested with an artificial event–station geometry using hard-assignment EMTomo and tomoDD with and without noise in its initial hypocenters; its matched soft-assignment comparison remains to be completed. Third, the same layered model is tested with event and station locations taken from the 2011 Geysers configuration, including a separate EMTomo soft-assignment candidate-count study. EMTomo requires no initial hypocenters, whereas tomoDD does; the methods also use different velocity parameterizations. We therefore evaluate tomoDD against a trilinear oracle fitted to the known velocity field as a reference for what its representation can resolve, rather than interpreting every difference from the true field as an inversion error. The comparisons characterize these specified setups, not a universal advantage of either method.

[TODO: Before submission: provide a stable repository link and a code-availability statement if the implementation is released. In Discussion, note that joint P/S inversion and time-lapse applications are outside the scope of this study. Remaining experiment-specific checks are listed at the end of Section 3.]

2 Method

This section describes EMTomo's v1.4 framework used in the layered-model experiments. The older checkerboard runs are identified separately in Section 3.3.

2.1 Parameterization and observations

We represent the isotropic velocity model by L values $v_\ell > 0$ at the centers of uniform cubic inversion cells, and perform the update in slowness $s_\ell = 1/v_\ell$. The vector $\mathbf{s} = (s_1, \ldots, s_L)^\mathsf{T}$ contains the slowness values of all inversion cells. Similar blockwise slowness models were used by Aki and Lee (1976). For travel-time calculations, each inversion cell is divided into equal cubic subcells; the subdivision factor sets their number along each side. This does not add inversion parameters. The true model used to generate synthetic data may have an even finer grid. Slowness on the computational grid can be assigned by nearest-neighbor or trilinear interpolation. We use nearest-neighbor interpolation in all experiments: each subcell has the slowness of its parent inversion cell. An initial velocity model is required.

For a catalog of E events, let t_{en} be the observed arrival time of one selected seismic phase from event e at station n. Its predicted value is $o_e + T_n(\mathbf{h}_e; \mathbf{s})$, where $\mathbf{h}_e \in \mathbb{R}^3$ is the hypocenter, o_e is the origin-time offset, and T_n is the travel time to station n. With Gaussian errors, origin time can be integrated out analytically, as in NonLinLoc (Lomax et al., 2000). Given the velocity model and the hypocenter, its estimate is

$$\widehat{o}_e = \frac{\sum_n p_{en}\left[t_{en} - T_n(\mathbf{h}_e; \mathbf{s})\right]}{\sum_n p_{en}}, \qquad p_{en} = \sigma_{en}^{-2}, \tag{1}$$

where σ_{en}^2 is the timing-error variance defined in Section 2.3. This estimate uses the same time reference as the observed arrivals. After eliminating origin times, the problem has $L + 3E$

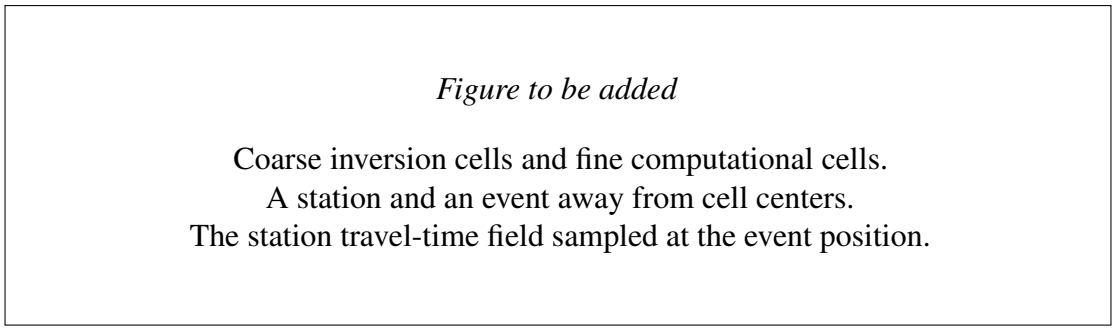


Figure 1: Inversion and computational grids, with travel-time evaluation for one station–event pair.

remaining unknowns: L slowness values and three coordinates per event. We do not require initial hypocenters, but we do need to know which event each arrival belongs to.

2.2 Travel-time fields on the fine grid

On the computational grid defined above, we use the second-order fast-marching solver in `scikit-fmm` (The scikit-fmm team, n.d.) to compute one first-arrival travel-time field $T_n(\mathbf{x}; \mathbf{s})$ per station at each iteration. By reciprocity, the travel time from a station to an event equals the time from the event to the station. We can therefore reuse each station field for all events.

Stations and events need not lie at grid centers. The field initialization uses the actual station coordinates, and trilinear interpolation gives travel times at continuous event coordinates. This interpolates travel times, not the blockwise slowness model. Figure 1 illustrates the grids and the travel-time calculation for one station–event pair.

2.3 Event-location step: candidate positions and weights

We keep the velocity model fixed and use the station travel-time fields computed in the previous step (Section 2.2). Each field gives the travel time between a station and each fine-grid cell center. For a trial event position $\mathbf{h}$, subtracting this travel time from the observed arrival gives a station-specific estimate of origin time:

$$\widehat{o}_{en}(\mathbf{h}) = t_{en} - T_n(\mathbf{h}; \mathbf{s}). \tag{2}$$

We do this for all stations that recorded the event. These estimates refer to the same event time, so we compare them with their weighted mean $\widehat{o}_e(\mathbf{h})$ from Eq. (1). We define the timing residual and the weighted squared misfit as

$$\delta t_{en}(\mathbf{h}) = \widehat{o}_{en}(\mathbf{h}) - \widehat{o}_e(\mathbf{h}), \qquad \chi^2_e(\mathbf{h}) = \sum_n p_{en}(\mathbf{h})\,[\delta t_{en}(\mathbf{h})]^2. \tag{3}$$

A smaller misfit means that the stations give more consistent estimates of origin time. The weights $p_{en} = 1/\sigma^2_{en}$ account for arrival-time uncertainty. We use

$$\sigma^2_{en}(\mathbf{h}) = \big(\alpha T_n(\mathbf{h};\mathbf{s})\big)^2 + \sigma^2_{\mathrm{pick}} + \sigma^2_{\mathrm{model}}, \qquad p_{en}(\mathbf{h}) = \sigma^{-2}_{en}(\mathbf{h}), \tag{4}$$

where α is dimensionless, while σ_{pick} and σ_{model} are in seconds. The term αT_n allows larger timing errors for longer travel times; σ_{pick} describes absolute picking error, and σ_{model} allows for modeling error. The errors are assumed Gaussian, but their variances need not be equal. These three parameters are set before inversion, not estimated during it. By default, α and σ_{pick} are read from the synthetic-data noise settings, unless explicitly overridden; σ_{model} is specified separately.

The squared misfit is the data-fit part of our criterion. Integrating out origin time under a uniform prior gives the full negative log likelihood

$$\Phi_e(\mathbf{h}) = \frac{1}{2}\chi^2_e(\mathbf{h}) + \frac{1}{2}\sum_n \log \sigma^2_{en} + \frac{1}{2}\log\left(\sum_n p_{en}\right). \tag{5}$$

up to a constant shared by all trial positions. The logarithmic terms are needed when the variances change with the trial position. If the variances are fixed, ranking positions by Φ_e is equivalent to ranking them by χ^2_e.

Evaluating Φ_e at all fine-grid cell centers gives a cost field for the event. We select a chosen number of low-cost cells, skipping cells too close to those already selected. The minimum separation is set in fine-grid cells using the largest coordinate difference (Chebyshev distance). We then refine the position within each selected cell by a local search for a lower cost, using trilinearly interpolated travel times. This distance constraint applies to the selected cells, not to the final continuous positions. After refinement, we rank the candidates again and keep a chosen

number of the best locations. For these locations $\mathbf{h}_{ej}$ with costs Φ_{ej}, the weights are

$$w_{ej} = \frac{\exp[-(\Phi_{ej} - \min_k \Phi_{ek})/\theta]}{\sum_k \exp[-(\Phi_{ek} - \min_i \Phi_{ei})/\theta]}, \tag{6}$$

where $\theta > 0$ controls how strongly the weights favor the best candidate; $\theta = 1$ gives normalized likelihood weights.

We may test many candidates but keep only some for the velocity update. Hard mode uses only the best one, with weight one. Soft mode uses the retained candidates with the weights above. Testing many candidates is therefore not the same as using soft mode; with one retained candidate, both modes are identical.

2.4 Velocity-update step: ray sensitivities and inversion

At fixed candidate hypocenters, first-order changes in travel time are approximated by $T_n(\mathbf{h}_{ej}; \mathbf{s} + \delta\mathbf{s}) \simeq T_n(\mathbf{h}_{ej}; \mathbf{s}) + \sum_{\ell=1}^{L} G_{en\ell}^{(j)} \delta s_\ell$. The row $G_{en\ell}^{(j)}$ is approximated by the length of the ray from $\mathbf{h}_{ej}$ to station n within inversion cell ℓ. We compute spatial gradients of the station travel-time fields on the fine grid and interpolate them along each ray. Starting at the candidate hypocenter, we follow the negative gradient toward the station, accumulate path lengths in fine cells, and sum them into the coarse inversion cells. Within the small homogeneous sphere used to initialize the station field, path length is assigned to the station cell, consistently with that local approximation. Rays that do not reach the station are excluded from the update.

For each event and retained hypothesis, let $\mathbf{r}_{ej}$ collect observed minus predicted arrivals at stations with successful rays, let G_{ej} collect their sensitivity rows, and set $p_{en} = 1/\sigma_{en}^2(\mathbf{h}_{ej})$. Rather than add origin-time increments as unknowns, we remove their effect with precision-weighted station means:

$$\widetilde{r}_{en}^{(j)} = r_{en}^{(j)} - \frac{\sum_m p_{em} r_{em}^{(j)}}{\sum_m p_{em}}, \qquad \widetilde{G}_{en:}^{(j)} = G_{en:}^{(j)} - \frac{\sum_m p_{em} G_{em:}^{(j)}}{\sum_m p_{em}}. \tag{7}$$

Only valid stations enter these sums. Holding locations and weights fixed during this step, the

linearized, damped least-squares update minimizes

$$\sum_{e,j} w_{ej} \sum_{n} p_{en} \left(\widetilde{r}_{en}^{(j)} - \widetilde{G}_{en:}^{(j)} \delta \mathbf{s} \right)^2 + \delta \mathbf{s}^{\mathsf{T}} D \, \delta \mathbf{s}, \tag{8}$$

where D is a diagonal damping matrix; its entries can be increased for poorly illuminated cells. Accordingly, the implementation accumulates

$$H = \sum_{e,j} w_{ej} \widetilde{G}_{ej}^{\mathsf{T}} P_{ej} \widetilde{G}_{ej}, \qquad \mathbf{b} = \sum_{e,j} w_{ej} \widetilde{G}_{ej}^{\mathsf{T}} P_{ej} \widetilde{\mathbf{r}}_{ej}, \qquad (H + D)\delta \mathbf{s} = \mathbf{b}, \tag{9}$$

with $P_{ej} = \mathrm{diag}(p_{en})$. We update the slowness by $\delta \mathbf{s}$, convert back to velocity, and optionally limit the fractional velocity change in each cell. Repeating the location and velocity steps for a prescribed number of cycles produces the final model.

2.5 Mathematical basis and limitations

Under independent Gaussian pick errors, Eq. (5) follows from marginalizing the origin time. At $\theta = 1$, Eq. (6) gives posterior weights on the retained hypotheses with equal prior probabilities, not a full spatial posterior. Soft assignment averages their quadratic data-fit contributions; hard assignment retains only the best hypothesis. A related approach is DeepGEM (Gao et al., 2021), which uses a normalizing flow to approximate the source posterior in variational EM for blind seismic tomography; here the approximation is a finite set of weighted locations.

For isotropic propagation, small perturbations, and no ray-path switching, Fermat's principle supports the frozen-ray slowness linearization. Weighted centering is an origin-time-only form of parameter elimination, related to the broader source-parameter separation of Pavlis and Booker (1980); it does not project out spatial hypocenter corrections. With locations, weights, and variances fixed, $H \succeq 0$, and strictly positive damping gives $H + D \succ 0$. The unclipped update therefore uniquely minimizes Eq. (8), reducing this quadratic objective relative to a zero update unless that update is already optimal.

This local result does not establish monotonic improvement of the nonlinear likelihood. The candidate set changes between iterations, ray sensitivities are approximate, and model-dependent variances are frozen in the M-step; tempering and step clipping introduce further departures

from exact EM. Thus the standard EM monotonicity guarantee (Dempster et al., 1977) does not directly apply. Regularization stabilizes the update but cannot resolve velocity–location ambiguities unsupported by the data.

3 Synthetic experiments and results

3.1 Synthetic arrivals and comparison protocol

Synthetic first-arrival times are generated independently of EMTomo using PyKonal 0.4.1 (White et al., 2020). The generator solves the Cartesian eikonal equation from the true event coordinates on a refined nodal grid, initializes off-grid sources with local straight-segment slowness integrals, and samples receivers by interpolating the regular factor $T(\mathbf{x})/\|\mathbf{x}-\mathbf{h}\|$. For the v1.4 layered runs, this differs from EMTomo's station-centered `scikit-fmm` fields, homogeneous source-sphere initialization, and direct interpolation of travel times. Separate solvers, grids, and source treatments reduce numerical inverse crime; both still approximate the same isotropic first-arrival problem, so this is not a test against independent wave physics.

Before referencing each event to its earliest observed arrival, we add independent Gaussian errors

$$t_{en}^{\text{noisy}} = T_{en}^{\text{true}} + 0.01 T_{en}^{\text{true}} Z_{en}^{(1)} + \sigma_a Z_{en}^{(2)}, \qquad Z_{en}^{(1)}, Z_{en}^{(2)} \sim \mathcal{N}(0,1), \tag{10}$$

with $\sigma_a = 0.05$ s for the checkerboards and 0.01 s for the layered models, using seed 42. Thus the pre-reference noise variance is $(0.01 T_{en}^{\text{true}})^2 + \sigma_a^2$, not the square of a sum of standard deviations. All event–station pairs are observed. EMTomo and tomoDD use the same saved picks within each experiment, but only tomoDD receives an initial hypocenter catalog. In what follows, a *zero-noise start* means true initial hypocenters for tomoDD, not noise-free arrival times. Inter-event differential times for tomoDD are formed from these picks; no additional waveform cross-correlation observations are supplied.

The methods retain their native velocity representations and regularization. Their iteration counts are not equated: the checkerboards use 12 EMTomo updates and three tomoDD iterations; the layered tests use eight or 12 EMTomo updates and three tomoDD iterations. We evaluate the

specified checkpoints, including iteration three of a tomoDD run that continued further, rather than silently substituting its final model. These are comparisons of the stated workflows, not controlled tests differing only in location-uncertainty treatment. A single noise seed is used; the results do not establish statistical superiority across realizations.

3.2 Quality metrics and representation oracles

Velocity fields. We evaluate the native reconstructed fields at the same physical points $\{\mathbf{x}_q\}_{q=1}^{Q}$, without converting one method to the other's parameterization. EMTomo is piecewise constant; tomoDD is evaluated by its native trilinear velocity interpolation, including the fixed exterior nodes. The principal measures are

$$\mathrm{RMSE}_v = \sqrt{\frac{1}{Q}\sum_q [\widehat{v}(\mathbf{x}_q) - v^{\mathrm{true}}(\mathbf{x}_q)]^2}, \qquad \mathrm{MAE}_v = \frac{1}{Q}\sum_q |\widehat{v}(\mathbf{x}_q) - v^{\mathrm{true}}(\mathbf{x}_q)|. \tag{11}$$

Whole-volume scores are accompanied by regional scores: interfaces versus block interiors for checkerboards, and individual spheres versus the surrounding medium for layered models. Native coarse-grid quality logs and travel-time residuals are not interchangeable with these field errors.

Hypocenters. For hard EMTomo and tomoDD, define $d_e = \|\widehat{\mathbf{h}}_e - \mathbf{h}_e^{\mathrm{true}}\|$. We report the catalog RMS $\sqrt{E^{-1}\sum_e d_e^2}$, median, and 95th percentile, with coordinate biases as diagnostics. For soft EMTomo we use the largest-weight hypothesis for the same point-location comparison and additionally score the entire discrete distribution. Probabilistic location methods distinguish optimal locations from their uncertainty (Lomax et al., 2000; Smith et al., 2022); to evaluate our weighted hypotheses without imposing a Gaussian shape or an arbitrary smoothing bandwidth, we choose the energy score (Gneiting & Raftery, 2007):

$$\mathrm{ES}_e = \sum_j w_{ej}\|\mathbf{h}_{ej} - \mathbf{h}_e^{\mathrm{true}}\| - \frac{1}{2}\sum_{j,k} w_{ej}w_{ek}\|\mathbf{h}_{ej} - \mathbf{h}_{ek}\|. \tag{12}$$

Lower mean ES is better; for a single hypothesis it equals the mean location distance, not the RMS. This proper score assesses probabilistic accuracy, but a low score alone does not

demonstrate calibrated credible regions. We also retain the weighted-mean location error and the spread around that mean; averaging separated modes can place the mean in an implausible location, so it does not replace the mode-based comparison.

All events receive equal weight in catalog statistics. The available EMTomo locations belong to the E-step immediately preceding the selected last velocity update, whereas velocity scores use the post-update model. They are explicitly labeled as such, not presented as a final relocation in that updated model. A final location-only pass is needed for fully synchronized velocity–hypocenter checkpoints.

Spherical anomalies. Let v_b denote the known layered background and $\delta v^{\mathrm{true}} = v^{\mathrm{true}} - v_b$. In addition to RMSE on the nonzero support S_k of each sphere, we measure its signed template-amplitude recovery:

$$a_k = \frac{\sum_{\mathbf{x}_q \in S_k} [\widehat{v}(\mathbf{x}_q) - v_b(\mathbf{x}_q)]\, \delta v^{\mathrm{true}}(\mathbf{x}_q)}{\sum_{\mathbf{x}_q \in S_k} [\delta v^{\mathrm{true}}(\mathbf{x}_q)]^2}. \tag{13}$$

For an otherwise correct background, $a_k = 0$ means absent contrast, $a_k = 1$ the true template amplitude, and $a_k < 0$ reversed contrast. This is a projection onto the true anomaly, not a peak-velocity ratio: displacement, shape distortion, and background errors also affect it. RMSE inside the sphere and outside all spheres distinguishes contrast recovery from local error and degradation of the background. No thresholded sphere segmentation is needed.

Oracles. For each representation, a truth-only oracle fits velocity values to the known field, without using arrivals or estimating hypocenters. For constant cells, the volume squared-error optimum is the arithmetic mean of true velocity within each cell, *not* the inverse of mean slowness. For tomoDD we use the saved least-squares trilinear truth fit with the same free nodes and fixed exterior values as the inversion. These references quantify representational error, not the resolution attainable from the available rays, and are distinct from an inversion with fixed true hypocenters. A whole-volume oracle is not a lower bound for every regional RMSE, and the difference between reconstruction and oracle RMSE is not an additive decomposition of inversion error.

The checkerboard truth is exactly representable by EMTomo's aligned constant cells, so its

oracle is the true field. This is *not* true for the layered sphere models on the coarser inversion grid: they require a separate cell-mean oracle. Because the two layered experiments share the same truth and grids, they share both representation oracles and initial native fields.

3.3 Checkerboards with varying anomaly size

The first series occupies $240 \times 120 \times 120$ km. Alternating cubes have velocities 4750 and 5250 m/s and sides 60, 40, or 30 km (respectively $4 \times 2 \times 2$, $6 \times 3 \times 3$, or $8 \times 4 \times 4$ cubes). The same 1200 synthetic events and 98 surface stations are used in all cases. EMTomo has $24 \times 12 \times 12$ constant 10-km cells; tomoDD has 3456 free interior velocity nodes at the same centers, with fixed exterior faces. Both start at 5000 m/s. EMTomo uses one active location hypothesis, 12 updates, subdivision nine, damping parameter 0.05, and a $\pm 0.5\%$ cellwise velocity-step limit. TomoDD uses three joint iterations, damping 150, smoothing 40/40/30, and initial hypocenters perturbed independently by 2 km standard deviation in each coordinate. Candidate initial coordinates outside the domain are rejection-resampled. Its event-pair graph uses a 30-km cutoff and maximum degree ten.

These saved checkerboard runs are labeled EMTomo v1.1 and predate the current unified likelihood and exact-coordinate station initialization. Their metadata does not preserve runtime source hashes; the archived pre-run source uses cell-center station seeding and an unweighted within-event pairwise residual, rather than the current precision-weighted formulation. We therefore treat this series as a legacy parameterization benchmark, not a validation of every implementation detail in Section 2.

All fields are scored at 432,000 common 2-km voxel centers. The initial-model RMSE is 250 m/s. Table 1 reports the reconstruction and oracle errors. The saved trilinear oracles were fitted on the original runs' sampling grids and then evaluated at these common points; they are representation references, not guaranteed exact minima of the common-grid objective.

TomoDD has lower volume RMSE for 60-km cubes; for 40- and 30-km cubes the differences favoring EMTomo are only 3.18 and 3.89 m/s. EMTomo has lower error within 5 km of true internal block faces in all three cases, whereas tomoDD has lower error in block interiors. The increasing trilinear-oracle error with decreasing cube size illustrates why sharper block bound-

Table 1: Checkerboard whole-volume velocity RMSE (m/s). EMTomo uses a single active location hypothesis. Its representation oracle has zero error in every case.

Cube side (km)	EMTomo	tomoDD	Trilinear oracle
60	185.62	142.30	76.10
40	166.15	169.33	97.24
30	169.33	173.22	116.37

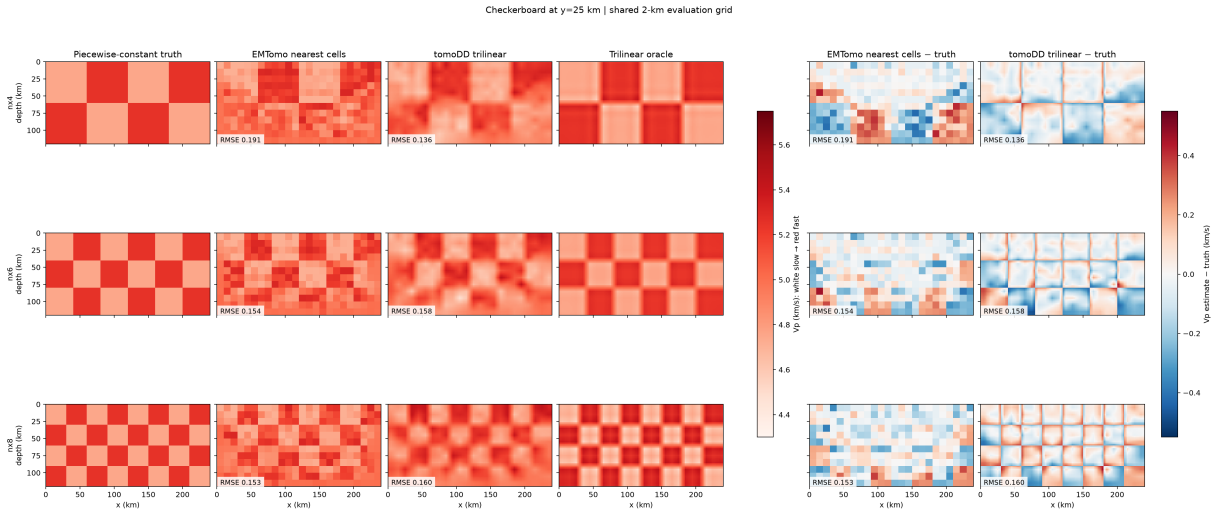


Figure 2: Checkerboard vertical section at $y = 25$ km. Rows have cube sides 60, 40, and 30 km; columns show truth, EMTomo, tomoDD, the trilinear oracle, and the two reconstruction errors. Figure velocity and error scales are in km/s; table scores are in m/s. Slice RMSE is not the whole-volume RMSE.

aries alone do not establish a resolution advantage (Figure 2). TomoDD's final hypocenter RMS is 1.774, 1.714, and 1.777 km, respectively, from the same 3.462-km initial RMS. EMTomo's corresponding location scores have not yet been extracted for this series.

3.3.1 Scaled checkerboards with the current implementation

Figures 3 and 4 show a separate, scaled checkerboard series on a $24 \times 12 \times 12$ km domain, with cube sides 6, 4, and 3 km ($n_x = 4, 6, 8$). The geometry contains 1200 events and 98 surface stations. Both methods start from a homogeneous 5 km/s velocity model and use 1-km velocity parameterizations. The saved observations in this series have 1% relative and 0.01-s absolute pre-reference noise. EMTomo uses 12 hard-assignment updates, 15 candidates with top-1 selection, regularization parameter $\lambda = 1$, and coverage-damping power zero. The tomoDD results are from three joint iterations with a 1-km per-coordinate Gaussian perturbation of the initial hypocenters (seed 42). The numerical scores in Table 1 describe the legacy, larger-domain

runs above, not these scaled reconstructions.

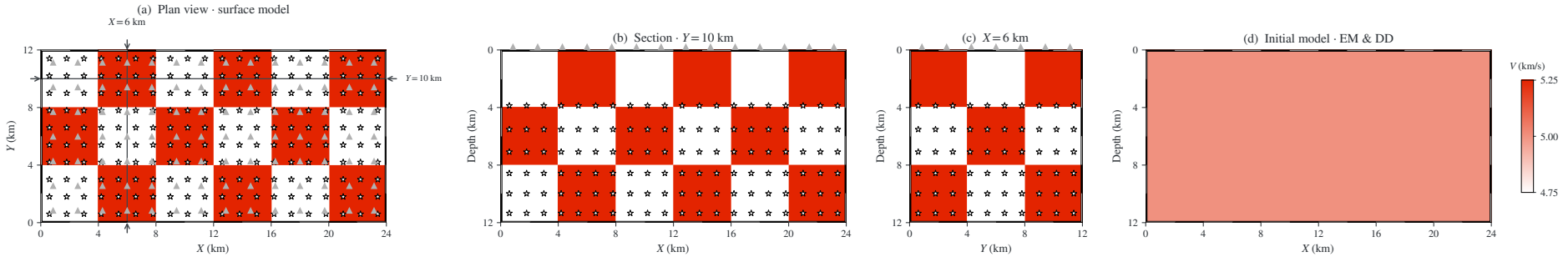


Figure 3: Scaled checkerboard model and acquisition geometry for $n_x = 6$. (a) Surface velocity field and plan-view projections of stations (grey triangles) and events (white stars); arrows mark the sections at $Y = 10$ km and $X = 6$ km. (b,c) The corresponding vertical velocity sections, with all stations and events projected onto each plane. Coincident projections are drawn once: each event marker represents six events in (a), ten in (b), and twenty in (c); station markers represent seven stations in (b) and fourteen in (c). (d) The common homogeneous initial velocity model for EMTomo and tomoDD, also used for $n_x = 4$ and 8. Depth is positive downward. Alternating frame segments are 2 km long.

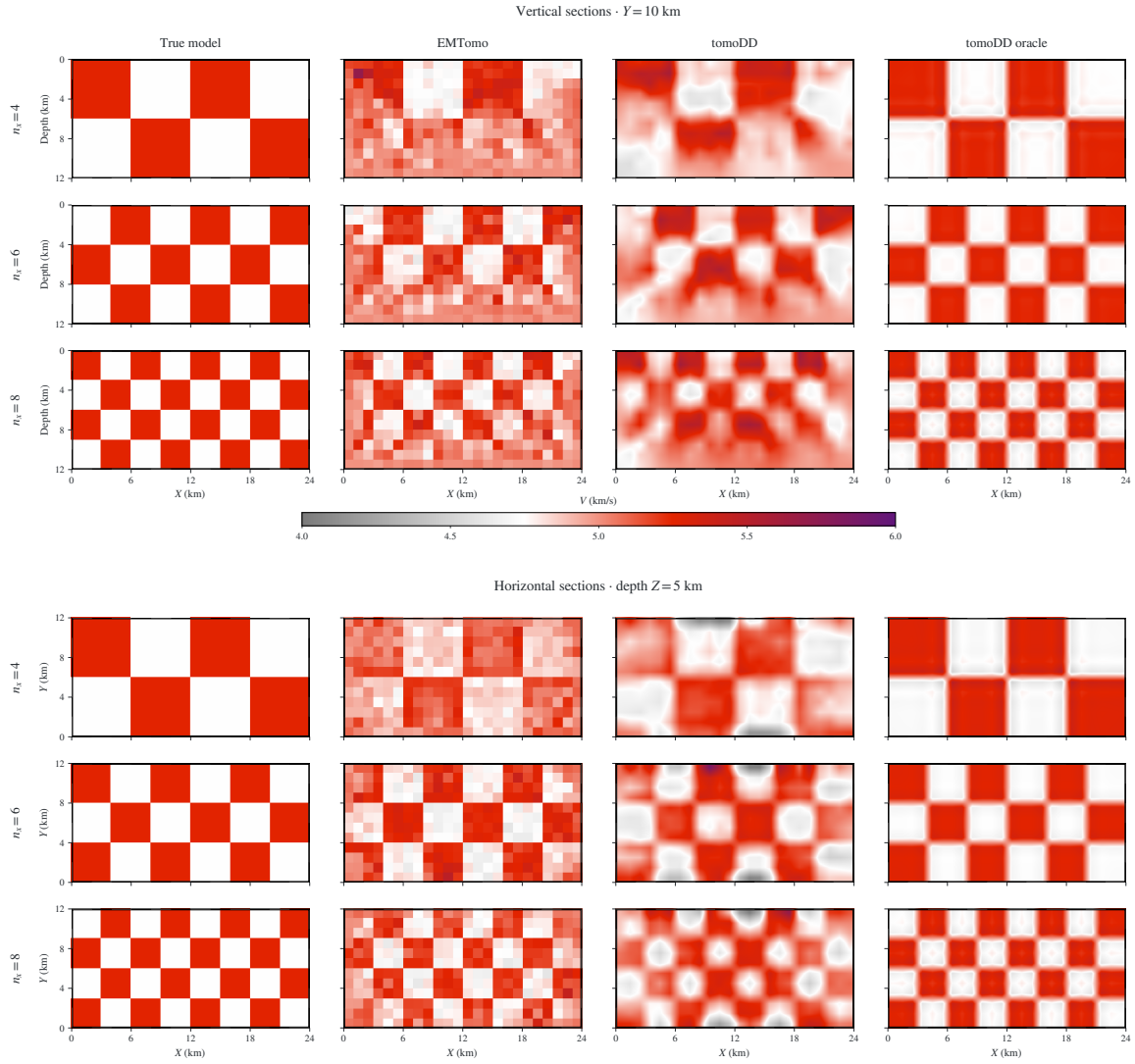


Figure 4: Scaled checkerboard reconstructions. Columns show truth, EMTomo after 12 updates with coverage-damping power zero, tomoDD after three iterations, and the saved trilinear representation oracle. The upper three rows are vertical sections at $Y = 10$ km for $n_x = 4, 6, 8$; the lower three rows are horizontal sections at depth $Z = 5$ km in the same order. True and EMTomo fields are piecewise constant; tomoDD and its oracle use native trilinear velocity interpolation. The oracle is a truth-only representation reference, not an inversion result. All panels share the velocity scale in km/s; no stations or events are overlaid.

Table 2: Spherical anomalies shared by the artificial- and Geysers-geometry experiments. Coordinates and radii are in km; peak contrast is in m/s. Depth is positive downward.

Sphere	x	y	z	Radius	Peak δv
1	3.5	2.5	1.2	0.9	$+600$
2	8.0	4.8	2.0	1.1	-650
3	12.5	2.4	2.6	1.2	$+700$
4	16.5	4.5	1.6	1.0	-600
5	6.0	2.2	3.6	1.1	$+700$
6	11.0	5.0	4.0	1.0	-650

3.4 Layered model with spherical anomalies: artificial geometry

The second series uses a $20 \times 7 \times 7$ km model sampled on 125-m truth cells. Four horizontal layers have velocities 3800, 4600, 5200, and 5700 m/s, with interfaces at 0.7, 1.9, and 3.4 km. Six additive spherical anomalies are specified in Table 2. Each has constant peak contrast inside half its radius and a cosine taper to zero at its outer radius. The starting model contains only the layers. Since the interfaces do not align with 1-km inversion cells, the starting layered field itself has representation error.

There are 1920 events on a $24 \times 8 \times 10$ lattice, with actual depths 0.75–5.25 km, and 52 stations on a 13×4 surface grid. EMTomo uses 980 constant 1-km cells, subdivision eight, and hard assignment after refining five candidate cells per event. We score its eight-update checkpoint (damping 0.5, coverage-damping exponent one, velocity-step limit $\pm 2\%$). TomoDD uses 980 free interior nodes with fixed padding, damping 40, smoothing 5/5/5, and a 0.05-km ray-derivative step. Its starts are either true hypocenters or locations perturbed by 0.5 km per coordinate; both are scored after three joint iterations. Their event-pair graphs use a 3-km cutoff and maximum degree four. No matched soft-EMTomo result is available yet for this geometry.

The common evaluation grid contains all 501,760 truth-cell centers. Only 15,314 centers belong to nonzero sphere support, so whole-volume error is dominated by the layered background. In the artificial geometry, EMTomo reduces volume RMSE from 215.43 to 202.86 m/s. Both tomoDD reconstructions recover the spheres more accurately in RMSE but have greater whole-volume error than their own trilinear starting field (Table 3). In this realization, a true initial catalog does not give the lowest tomoDD error. This cannot be attributed solely to initial-location accuracy: perturbing the catalog also changes the event-pair graph. Figure 5 shows the same

Table 3: Layered-model velocity RMSE (m/s), scored on the same 125-m truth grid. Zero and noisy tomoDD starts refer to initial hypocenters, not picks. The native initial fields and oracles are shared by both geometries. Regional oracle errors are descriptive, not regional lower bounds.

Geometry	Model	Volume	Spheres	Outside spheres
Artificial	EMTomo hard, 8 updates	202.86	327.62	197.66
	tomoDD, zero start	251.62	287.36	250.41
	tomoDD, noisy start	208.12	273.49	205.72
Geysers	EMTomo hard, 12 updates	208.32	329.69	203.32
	tomoDD, zero start	188.75	276.76	185.30
	tomoDD, noisy start	214.69	281.31	212.25
Both	EMTomo initial field	215.43	427.24	205.24
	tomoDD initial field	173.25	406.87	160.46
	Constant-cell oracle	181.49	298.06	176.58
	Trilinear oracle	142.86	233.91	139.03

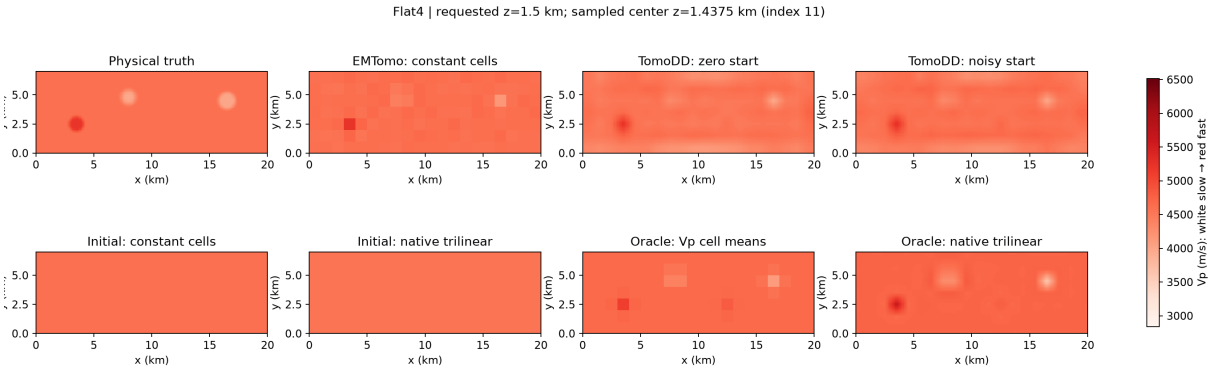


Figure 5: Layered model with artificial geometry, at the sampled truth-cell depth $z = 1.4375$ km (requested plane: 1.5 km). Initial models contain no spheres; their native interpolation and their representation oracles differ. No display smoothing is added.

physical section for the reconstructions, native initial models, and oracles.

The available EMTomo E-step locations have 147.74-m RMS error, compared with 258.58 and 222.08 m for the two selected tomoDD catalogs (Table 4). The noisy tomoDD start has 852.30-m initial RMS; the zero start initially has exactly zero location error. Thus relocation accuracy and velocity accuracy need not rank the workflows in the same order.

3.5 Same velocity model with the 2011 Geysers geometry

The third series changes the acquisition geometry while keeping the velocity truth, inversion grids, and pick-noise model fixed. It uses 3776 catalog event locations from 1–20 April 2011 and 26 Geysers station positions; all arrival times remain synthetic. The model top is placed at

Table 4: Full-catalog location errors (m). EMTomo locations are from the E-step preceding the last scored velocity update, not a relocation in the post-update model. TomoDD catalogs are from iteration three.

Geometry	Catalog	3D RMS	Median	P95
Artificial	EMTomo hard	147.74	107.78	264.59
	tomoDD, zero start	258.58	216.65	456.58
	tomoDD, noisy start	222.08	184.18	394.76
Geysers	EMTomo hard	145.96	113.78	262.06
	tomoDD, zero start	141.90	106.11	255.23
	tomoDD, noisy start	180.58	123.33	359.83

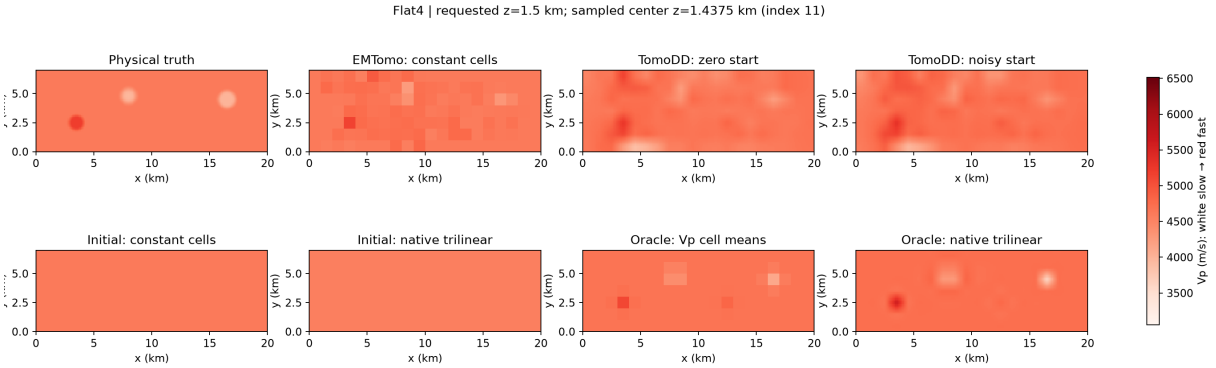


Figure 6: The same layered truth and $z = 1.4375$ km section as Figure 5, with Geysers event–station geometry. These are reconstructions from synthetic picks, not an inversion of observed Geysers arrivals.

elevation 1166 m, giving station depths 0–839 m and event depths 1.166–6.877 km. Four events outside the transformed domain are omitted. Station elevations and catalog depths are assumed to share a vertical datum, which has not been independently verified; no terrain or air layer is modeled.

EMTomo hard uses one candidate per event, 12 updates, damping one, and coverage-damping exponent two; other grid and step settings match the artificial-geometry test. TomoDD retains the same three-iteration protocol and zero/0.5-km initial-location perturbations. Whole-volume RMSE is 208.32 m/s for EMTomo and 188.75/214.69 m/s for zero/noisy-start tomoDD. Both tomoDD runs again improve sphere RMSE relative to initialization but worsen volume RMSE relative to their own initial trilinear field. The zero-start tomoDD and EMTomo location RMS values are close (141.90 and 145.96 m); the noisy-start tomoDD value is 180.58 m, from 868.86 m initially. Figure 6 uses the same section and scale as Figure 5.

Template amplitudes (Table 5) show that aggregate sphere RMSE does not describe every recovered anomaly. For example, EMTomo's amplitude for sphere three is 0.746 with artificial

Table 5: Signed sphere-template amplitudes a_k from Eq. (13). One denotes the true template amplitude. Both oracles are fitted over the whole volume, not separately to maximize each sphere's contrast.

Model	a_1	a_2	a_3	a_4	a_5	a_6
Artificial: EMTomo hard	.439	.617	.746	.784	.525	.323
Artificial: tomoDD zero	.174	.610	.768	.815	.780	.305
Artificial: tomoDD noisy	.346	.551	.840	.722	.817	.281
Geysers: EMTomo hard	.443	.533	.827	.547	.602	.271
Geysers: tomoDD zero	.516	.448	.834	.476	.958	.428
Geysers: tomoDD noisy	.652	.373	.870	.341	.956	.508
Constant-cell oracle	.485	.503	.677	.753	.497	.463
Trilinear oracle	.666	.677	.908	.862	.741	.627

Table 6: Geysers candidate-count sensitivity. Velocity errors are in m/s; all location quantities are in m. Location scores use the preceding E-step for every run.

Assignment	Volume RMSE	Sphere RMSE	MAP RMS	Mean ES	RMS spread
Hard, $K = 1$	208.32	329.69	145.96	127.68	0.00
Soft, $K = 5$	208.27	330.13	140.12	103.36	55.72
Soft, $K = 10$	208.26	330.20	138.30	95.38	80.31
Soft, $K = 15$	208.29	330.18	138.65	93.24	94.32

geometry and 0.827 with Geysers geometry, whereas sphere six reaches only 0.323 and 0.271. The native cell-mean oracle itself yields amplitudes below one, so under-recovered contrast is partly a representation effect. A reconstruction can exceed an oracle's individual template amplitude without improving its whole-volume error.

3.5.1 Soft-assignment sensitivity

Separate completed Geysers runs retain $K = 5, 10, 15$ hypotheses at temperature one. They share the hard run's input hashes, initial velocity field, regularization, and 12-update endpoint. However, both the shortlist size and retained-hypothesis count change with K; comparison with hard $K = 1$ is not an isolated test of soft versus hard weights. Table 6 reports every available K, without selecting the best result post hoc.

The available soft runs have lower mean energy scores and slightly lower mode-location RMS, but essentially unchanged volume velocity RMSE and slightly higher sphere RMSE. Increasing the number of retained hypotheses therefore does not produce an appreciable velocity-reconstruction gain in this example. RMS spread in the table is the catalog root-mean weighted squared distance about each event's weighted mean, not a calibrated confidence radius.

[TODO: Verify the legacy checkerboard runtime implementation or repeat the series with the current implementation; complete the matched soft/hard experiment for artificial geometry; extract checkerboard EMTomo location scores; perform final location-only passes to synchronize EMTomo velocity and location checkpoints. For a clean weight-treatment ablation, keep the same candidate shortlist in hard and soft runs. Confirm the Geysers catalog/station vertical datum and document checkpoint-selection rationale and runtimes before submission.]

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